

SOME BOUNDS FOR EXCEEDANCE PROBABILITIES WITH APPLICATIONS

J.H. Venter

Potchefstroom University

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1. Introduction

Let $\{T_n\}$ be a sequence of random variables and $\{c_n\}$ a sequence of real numbers; the problem studied in this note is that of finding upper bounds for probabilities of the form

$$(1) \quad P(T_n \geq c_n \text{ for some } n \in [m, \ell])$$

with m and ℓ integers, $m < \ell$.

Such results find application in certain areas of sequential analysis, particularly the calculation of confidence sequences and selection procedures. Some applications are indicated in paragraphs 3 and 4 while paragraph 2 contains the main result which essentially gives a method by means of which one can obtain a more refined bound for (1) from a possibly crude initial bound.

2. Main Result

The following results are to some extent motivated by the analysis of Darling and Robbins (1967a).

Theorem.

Let m and ℓ be integers with $m \leq \ell$; suppose that for all integers r and s with $m \leq r \leq s \leq \ell$ we have the bound

$$(2) \quad P(T_n \geq c_n \text{ for some } n \in [r, s]) \leq a(r, s).$$

Then we also have

$$(3) \quad P(T_n \geq c_n \text{ for some } n \in [m, \ell]) \leq b(m, \ell)$$

where

$$(4) \quad b(m, \ell) = \min_{n \in [m, \ell]} \{b(m, n-1) + a(n, \ell)\}$$

and $b(m, m-1)$ has to be taken as zero.

Proof:

Let $m = m_0 < m_1 < \dots < m_p = \ell + 1$; then the lefthand side of (3) is bounded by

$$(5) \quad \sum_{i=0}^{p-1} P(T_n \geq c_n \text{ for some } n \in [m_i, m_{i+1} - 1]) \\ \leq \sum_{i=0}^{p-1} a(m_i, m_{i+1} - 1) = B(m, \ell, M)$$

say, where M denotes the partition

$$\cup_{i=0}^{p-1} [m_i, m_{i+1}) \text{ of } [m, \ell]. \text{ Put}$$

$$(6) \quad b(m, \ell) = \min_M B(m, \ell, M).$$

Then (3) holds and it is clear that (4) also holds for $\ell = m$. Suppose that $\ell > m$ and let $M_{m,\ell}^*$ denote a partition which minimizes $B(m, \ell, M)$. Suppose that $M_{m,\ell}^*$ is the partition corresponding to $\cup_{i=0}^{p-1} [m_i, m_{i+1})$; denote by M the partition of

$$[m, m_{p-1}) = \cup_{i=0}^{p-2} [m_i, m_{i+1}); \text{ we then have}$$

$$\begin{aligned} B(m, \ell, M_{m,\ell}^*) &= B(m, m_{p-1} - 1, \bar{M}) + a(m_{p-1}, \ell) \\ &\geq b(m, m_{p-1} - 1) + a(m_{p-1}, \ell) \\ &\geq \min_{n \in [m, \ell]} \{b(m, n-1) + a(n, \ell)\} \\ &= b(m, n_0 - 1) + a(n_0, \ell) \end{aligned}$$

with $n_0 \in [m, \ell]$. The last sum equals $B(m, \ell, M_0)$ where M_0 is obtained from M_{m, n_0-1}^* by addition of $[n_0, \ell]$. However, $B(m, \ell, M_0) \geq B(m, \ell, M_{m,\ell}^*)$ by definition; hence equalities hold above and the theorem follows.

Remarks:

1. It is clear that $b(m, \ell) \leq a(m, \ell)$; the theorem therefore gives a recursive method by means of which one can proceed from the initial (possibly crude) bound a to the sharper bound b .
2. We list some ways in which the initial bounds a can be obtained.

2.1 Suppose that for constant c we have the bound

$$(7) \quad P(T_n \geq c \text{ for some } n \in [r, s]) \leq a_1(c, r, s).$$

Then, if $\{c_n\}$ is non-decreasing we have

$$\begin{aligned} & P(T_n \geq c_n \text{ for some } n \in [r, s]) \\ & \leq P(T_n \geq c_r \text{ for some } n \in [r, s]) \\ & \leq a_1(c_r, r, s) \end{aligned}$$

so that we can take

$$(8) \quad a(r, s) = a_1(c_r, r, s).$$

Again, if $\{c_n\}$ is non-increasing we can, by similar reasoning, take

$$(9) \quad a(r, s) = a_1(c_s, r, s).$$

2.2 Suppose that for constants α, β we have the bounds

$$(10) \quad P(T_n \geq \alpha) \leq a_2(\alpha, n)$$

$$(11) \quad P(T_n \geq \alpha + \beta n \text{ for some } n \in [r, s]) \leq a_3(\alpha, \beta, r, s).$$

Then, if $\{c_n\}$ is concave in n we have for $n \in [r, s]$, $r < s$,

$$(12) \quad c_n \geq ((s-n)c_r + (n-r)c_s)/(s-r)$$

and hence, for $r < s$,

$$\begin{aligned} & P(T_n \geq c_n \text{ for some } n \in [r, s]) \\ & \leq a_3((sc_r - rc_s)/(s-r), (c_s - c_r)/(s-r), r, s). \end{aligned}$$

Consequently we can then take the bound $a(r, s)$ of (2) to be

$$(13) \quad a(r, s) = \begin{cases} a_2(c_r, r) & \text{if } r = s \\ a_3((sc_r - rc_s)/(s-r), (c_s - c_r)/(s-r), r, s) & \text{if } r < s. \end{cases}$$

Again, if $\{c_n\}$ is convex in n , we have, for $k \in [r, s]$ and $n \in [r, s]$

$$(14) \quad c_n \geq (k+1-n)c_k + (n-k)c_{k+1}$$

and hence, for $r < s$,

$$P(T_n \geq c_n \text{ for some } n \in [r, s]) \\ \leq \min_{k \in [r, s]} a_3((k+1)c_k - kc_{k+1}, c_{k+1} - c_k, r, s)$$

so that we may then take

$$(15) \quad a(r, s) = \begin{cases} a_2(c_r, r) & \text{if } r = s \\ \min_{k \in [r, s]} a_3((k+1)c_k - kc_{k+1}, c_{k+1} - c_k, r, s) & \text{if } r < s. \end{cases}$$

3. Application to sums and means

Let $\{X_n\}$ be independent and identically distributed random variables with

$$(16) \quad E \exp(tX_i) = \phi(t) \leq \exp(\frac{1}{2}t^2)$$

for all real t . We want to obtain bounds for

$$(17) \quad P(S_n \geq Bn^{1/2}(\log \log n + d)^{1/2} \text{ for some } n \in [m, \ell])$$

where $S_n = \sum_{i=1}^n X_i$ and $B, d > 0$. Interest in this type of probability stems from the law of the iterated logarithm (cf. Robbins and Darling (1967b), (1967c)). Our results can be used in various ways for this purpose.

In the first place with

$$(18) \quad c_n = Bn^{1/2}(\log \log n + d)^{1/2}$$

we have $\{c_n\}$ concave for $n \geq 2$; from the results of Khan (1973) we find that (11) holds with T_n replaced by S_n and

$$(19) \quad a_3(\alpha, \beta, r, s) = \begin{cases} \exp\{-\frac{1}{2}(\alpha+\beta r)^2/r\} & \text{if } \alpha \leq r\beta \\ \exp\{-2\alpha\beta\} & \text{if } \alpha > r\beta. \end{cases}$$

Also, from Bernstein's inequality we get (10) with

$$(20) \ a_2(\alpha, n) = \exp(-\frac{1}{2}\alpha^2/n).$$

Taking $a(r, s)$ as in (13) with these choices of a_2 and a_3 we can now apply the theorem to obtain the bounds b .

Some examples of numerical results obtained in this way follow:

1. With $B = 2,12$, $d = 1,236$, $m = 16$ and $\ell = \infty$ Darling & Robbins (1967b) find that (17) is less than 0,05. Using the above method for these choices of B , d and m we have obtained the following values for $b(16, \ell)$:

ℓ	100	250	500
$b(16, \ell)$	0,0101	0,0117	0,0129

2. With $B = 2,12$, $d = 1,457$, $m = 32$ and $\ell = \infty$, Darling & Robbins (1967c) find the bound 0,025; with our method we get

ℓ	100	250	500
$b(32, \ell)$	0,0029	0,0043	0,0052

In both these cases the limiting value of $b(m, \ell)$ as $\ell \rightarrow \infty$ should be substantially below the values of Darling and Robbins.

Another approach towards obtaining initial bounds for (17) is this:

In Darling and Robbins (1967a) it is shown that

$$P[\bar{X}_n \geq c \text{ for some } n \geq r] \leq \exp(-\frac{1}{2}rc^2).$$

This result can also be obtained from the fact that $\{\bar{X}_n\}$ constitutes a reversed martingale and use of the Bernstein inequality for reversed martingales. With $c_n = Bn^{\frac{1}{2}}(\log \log n + d)^{\frac{1}{2}}$ we have c_n decreasing in n : hence we can take $a(r, s)$ as in (9), viz. $a(r, s) = \exp(-\frac{1}{2}rc_s^2)$. The bounds obtained in this way for the numerical cases mentioned above are however not as good as those given there.

4. Application to the one sample rank statistic

Let $X_1, X_2, \dots$ denote a sequence of iid random variables symmetrically

distributed about 0 with distribution function F . Let

$$(21) \quad S_n = \sum_{i=1}^n J_n(i/(n+1))(Z_{in} - \frac{1}{2})$$

where $Z_{in} = 1$ if the X_j corresponding to the i -th smallest under the first n absolute X_j 's is positive and $Z_{in} = 0$ otherwise. The score function J_n and the distribution function F are assumed to satisfy the usual conditions (see e.g. Sen and Ghosh (1971)).

With $A_n^2 = n^{-1} \sum J_n^2(i/(n+1))$ it is shown in Steyn and Geertsema (1974) that

$$(22) \quad E \exp(tS_n) \leq \exp(nA_n^2 t^2/8)$$

while $\{S_n\}$ constitutes a martingale sequence according to Theorem 4.5 of Sen and Ghosh (1971). From Bernstein's inequality for martingales we get

$$(23) \quad P(S_n \geq c \text{ for some } n \in [r, s]) \leq \exp(-2c^2/sA_s^2).$$

Motivated again by a law of the iterated logarithm (Sen and Ghosh (1973)) one might here be interested in

$$(24) \quad P(S_n \geq c_n \text{ for some } n \in [m, l])$$

with either

$$(25) \quad c_n = BA_n^{1/2}(\log \log n + d)^{1/2} \quad .$$

or

$$(26) \quad c_n = BA_n^{1/2}(\log \log n + d)^{1/2}$$

with $A^2 = \lim A_n^2$ and $B, d > 0$. $\{c_n\}$ as defined by (26) is increasing and this is sometimes also the case for c_n as given by (25). For example, in the Wilcoxon case $J_n(u) = u$ and $A_n = ((2n+1)/6(n+1))^{1/2}$ so that c_n of (25) is indeed increasing. When this is true it follows from (23) that we can take

$$(27) \quad a(r, s) = \exp(-2 c_r^2/sA_s^2)$$

as in (8) and apply our theorem. The results can then be used for the construction of confidence sequences as in Steyn and Geertsema (1974); the intervals obtained for a given confidence are generally of shorter length.

Some idea of the sharpness of these bounds can be obtained by means of the results

of Miller (1972). He gives approximate values for constants w_N^α such that

$$(28) \quad P(SR_n \geq nw_N^\alpha \text{ for some } n \in [1, N]) = \alpha$$

where his statistic SR_n is related to S_n above by $SR_n = 2(n+1)S_n$ for the Wilcoxon case $J_n(u) = u$. (28) can then be written

$$(29) \quad P(S_n \geq c_n \text{ for some } n \in [1, N]) = \alpha \text{ where}$$

$$(30) \quad c_n = \frac{1}{2}nw_N^\alpha/(n+1)$$

which is increasing in n . We can then take the bound a as in (27) with this new choice of c_n and compute $b(1, N)$ from our theorem and compare with α .

The following numerical results have been obtained:

α	0,01		0,05		0,10	
w_{50}^a	10,73	10,52	7,82	8,00	6,61	6,72
$b(1,50)$	0,041	0,047	0,199	0,183	0,331	0,317

The two cases given for each a correspond to the two approximations for w_{50}^α given by Miller (1972). It appears that the bound is about three to four times above the actual value in this case. From computations at smaller N values it appears that there is a tendency for the sharpness of the bounds to improve with increasing N .

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